

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{18} (-g_1^2 g_2^4 + g_2^6) \text{LF}_{3,0}[m_2] + \frac{1}{12} (-g_1^2 g_2^4 + g_2^6) \text{LF}_{4,-1}[m_2] + \frac{4}{45} g_2^4 (g_1^2 - g_2^2) \text{LF}_{5,-2}[m_2] - \right. \\
& \frac{1}{72} \sum_p g_1^6 \text{LF}_{3,0}[m_p^P] + \frac{1}{12} g_1^4 \overline{y_d^{PR}} y_d^{PR} \text{LF}_{3,0}[m_d^r] - \frac{1}{8} \sum_p g_1^6 \text{LF}_{3,0}[m_p^P] + \\
& \frac{1}{4} g_1^4 \overline{y_e^{PR}} y_e^{PR} \text{LF}_{3,0}[m_e^r] + \frac{1}{288} (18 \overline{y_e^{PR}} y_e^{PR} (g_1^2 + g_2^2)^2 + \sum_p (9 g_1^6 + g_1^2 g_2^4 - 10 g_2^6)) \\
& \text{LF}_{3,0}[m_l^P] + \frac{1}{96} \sum_p g_2^4 (-g_1^2 + g_2^2) \text{LF}_{4,-1}[m_l^P] + \\
& \frac{1}{45} \sum_p g_2^4 (-g_1^2 + g_2^2) \text{LF}_{5,-2}[m_l^P] + \frac{1}{288} (6 \overline{y_d^{PR}} y_d^{PR} (g_1^2 - 3 g_2^2)^2 - \\
& 12 \overline{y_u^{PR}} y_u^{PR} (g_1^4 + 18 g_1^2 g_2^2 - 27 g_2^4) - \sum_p (g_1^6 - 39 g_1^2 g_2^4 + 30 g_2^6)) \text{LF}_{3,0}[m_q^P] + \\
& \frac{1}{32} g_2^2 (-g_1^2 + g_2^2) (-24 \overline{y_u^{PR}} y_u^{PR} + \sum_p g_2^2) \text{LF}_{4,-1}[m_q^P] + \frac{1}{15} \sum_p g_2^4 (-g_1^2 + g_2^2) \text{LF}_{5,-2}[m_q^P] + \\
& \frac{1}{9} \sum_p g_1^6 \text{LF}_{3,0}[m_p^P] - \frac{2}{3} g_1^4 \overline{y_u^{PR}} y_u^{PR} \text{LF}_{3,0}[m_u^r] + \frac{1}{36} (-g_1^2 g_2^4 + g_2^6) \text{LF}_{3,0}[\tilde{\mu}] + \\
& \frac{1}{24} (-g_1^2 g_2^4 + g_2^6) \text{LF}_{4,-1}[\tilde{\mu}] + \frac{2}{45} g_2^4 (g_1^2 - g_2^2) \text{LF}_{5,-2}[\tilde{\mu}] - \\
& \frac{1}{4} g_1^6 \text{LF}_{3,3,-3}[m_l, \tilde{\mu}] - \frac{3}{4} g_1^6 m_l^2 \text{LF}_{3,3,-2}[m_l, \tilde{\mu}] + \frac{1}{3} g_2^4 (g_1^2 - g_2^2) \text{LF}_{2,1,0}[m_2, \tilde{\mu}] + \\
& \frac{17}{24} g_2^4 (g_1^2 - g_2^2) \text{LF}_{2,2,-1}[m_2, \tilde{\mu}] + \frac{1}{6} (-g_1^2 g_2^4 + g_2^6) \text{LF}_{3,1,-1}[m_2, \tilde{\mu}] + \\
& \frac{7}{6} g_2^4 (-g_1^2 + g_2^2) \text{LF}_{3,2,-2}[m_2, \tilde{\mu}] + \frac{3}{4} g_2^4 m_2^2 (g_1^2 - g_2^2) \text{LF}_{3,2,-1}[m_2, \tilde{\mu}] + \\
& \frac{1}{4} g_2^4 (2 g_1^2 - 11 g_2^2) \text{LF}_{3,3,-3}[m_2, \tilde{\mu}] - \frac{1}{4} g_2^4 m_2^2 (2 g_1^2 + g_2^2) \text{LF}_{3,3,-2}[m_2, \tilde{\mu}] + \\
& \frac{1}{2} g_2^4 (g_1^2 - g_2^2) \text{LF}_{4,2,-3}[m_2, \tilde{\mu}] + \frac{1}{2} g_2^4 m_2^2 (-g_1^2 + g_2^2) \text{LF}_{4,2,-2}[m_2, \tilde{\mu}] + \\
& \frac{1}{12} (g_1^2 \overline{a_d^{PR}} a_d^{PR} (2 g_1^2 - 3 g_2^2) + 2 g_1^4 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR}) \text{LF}_{2,2,0}[m_d^r, m_q^P] + \\
& \frac{1}{12} g_1^4 (\overline{a_d^{PR}} a_d^{PR} - 2 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR}) \text{LF}_{3,1,0}[m_d^r, m_q^P] + \frac{1}{2} g_1^2 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} \text{LF}_{2,1,0}[m_d^r, m_u^t] + \\
& \frac{1}{4} g_1^2 (-g_2^2 \overline{a_e^{PR}} a_e^{PR} + 2 g_1^2 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR}) \text{LF}_{2,2,0}[m_e^r, m_l^P] + \\
& \frac{1}{4} g_1^4 (\overline{a_e^{PR}} a_e^{PR} - 2 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR}) \text{LF}_{3,1,0}[m_e^r, m_l^P] + \\
& \frac{1}{4} (g_2^4 \overline{a_e^{PR}} a_e^{PR} - 2 g_1^4 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR}) \text{LF}_{3,1,0}[m_l^P, m_e^r] + \\
& \frac{1}{8} (g_1^2 \overline{a_e^{PR}} a_e^{PR} (-g_1^2 + g_2^2) - 2 g_1^4 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR}) \text{LF}_{3,2,-1}[m_l^P, m_e^r] + \\
& \frac{3}{8} (g_2^2 \overline{a_e^{PR}} a_e^{PR} (g_1^2 - g_2^2) + 2 g_1^4 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR}) \text{LF}_{4,1,-1}[m_l^P, m_e^r] + \\
& \frac{1}{24} (g_1^2 - g_2^2) (3 \overline{a_e^{PR}} a_e^{PR} (g_1^2 - g_2^2) - 2 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR} (3 g_1^2 + g_2^2)) \text{LF}_{5,1,-2}[m_l^P, m_e^r] + \\
& \frac{1}{12} (\overline{a_d^{PR}} a_d^{PR} (2 g_1^2 - 3 g_2^2)^2 - 2 g_1^4 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR}) \text{LF}_{3,1,0}[m_q^P, m_d^r] + \\
& \frac{1}{8} (g_1^2 \overline{a_d^{PR}} a_d^{PR} (-g_1^2 + g_2^2) - 2 g_1^4 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR}) \text{LF}_{3,2,-1}[m_q^P, m_d^r] - \\
& \frac{3}{8} (\overline{a_d^{PR}} a_d^{PR} (2 g_1^4 - 5 g_1^2 g_2^2 + 3 g_2^4) - 2 g_1^4 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR}) \text{LF}_{4,1,-1}[m_q^P, m_d^r] + \\
& \frac{1}{8} (g_1^2 - g_2^2) (3 \overline{a_d^{PR}} a_d^{PR} (g_1^2 - g_2^2) - 2 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR} (3 g_1^2 + g_2^2)) \text{LF}_{5,1,-2}[m_q^P, m_d^r] + \\
& \frac{1}{4} \overline{y_u^{ST}} y_u^{PT} (g_1^2 - 3 g_2^2) (\overline{y_d^{PR}} y_d^{SR} + 2 \overline{y_u^{PR}} y_u^{SR}) \text{LF}_{2,1,0}[m_q^P, m_q^S] + \\
& \frac{3}{4} \overline{y_u^{PR}} \overline{y_u^{ST}} y_u^{PT} y_u^{SR} (-g_1^2 + g_2^2) \text{LF}_{3,1,-1}[m_q^P, m_q^S] + \\
& \frac{1}{48} (2 \overline{a_u^{PR}} a_u^{PR} (7 g_1^4 - 6 g_1^2 g_2^2 + 15 g_2^4) - \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (g_1^2 + 3 g_2^2)^2) \text{LF}_{2,2,0}[m_q^P, m_u^r] + \\
& \frac{1}{48} (-2 \overline{a_u^{PR}} a_u^{PR} (g_1^4 + 18 g_1^2 g_2^2 - 27 g_2^4) + \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (g_1^2 + 3 g_2^2)^2) \text{LF}_{3,1,0}[m_q^P, m_u^r] + \\
& \frac{1}{2} g_2^2 \overline{a_u^{PR}} a_u^{PR} (g_1^2 - g_2^2) \text{LF}_{3,2,-1}[m_q^P, m_u^r] + \frac{3}{4} g_2^2 \overline{a_u^{PR}} a_u^{PR} (g_1^2 - g_2^2) \text{LF}_{4,1,-1}[m_q^P, m_u^r] + \\
& \frac{1}{4} \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} (g_1^2 - 3 g_2^2) \text{LF}_{2,1,0}[m_q^S, m_q^P] + \\
& \frac{1}{48} (2 \overline{a_u^{PR}} a_u^{PR} (-49 g_1^4 + 6 g_1^2 g_2^2 + 3 g_2^4) + \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (g_1^2 + 3 g_2^2)^2) \text{LF}_{3,1,0}[m_u^r, m_q^P] + \frac{1}{16} \\
& (-2 \overline{a_u^{PR}} a_u^{PR} (g_1^4 - 2 g_1^2 g_2^2 + 5 g_2^4) - \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (g_1^4 + 2 g_1^2 g_2^2 - 3 g_2^4)) \text{LF}_{3,2,-1}[m_u^r, m_q^P] + \\
& \frac{3}{16} (2 \overline{a_u^{PR}} a_u^{PR} (7 g_1^4 - 2 g_1^2 g_2^2 - g_2^4) + \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (g_1^4 + 2 g_1^2 g_2^2 - 3 g_2^4)) \text{LF}_{4,1,-1}[m_u^r, m_q^P] - \\
& \frac{1}{8} (g_1^2 - g_2^2) (2 \overline{a_u^{PR}} a_u^{PR} (3 g_1^2 + g_2^2) + 3 \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (-g_1^2 + g_2^2)) \text{LF}_{5,1,-2}[m_u^r, m_q^P] + \\
& g_1^2 \overline{y_u^{PR}} \overline{y_u^{ST}} y_u^{PT} y_u^{SR} \text{LF}_{2,1,0}[m_u^r, m_u^t] - \frac{1}{2} \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} (2 g_1^2 + 3 g_2^2) \text{LF}_{2,1,0}[m_u^t, m_d^r] + \\
& \frac{3}{2} g_2^2 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} \text{LF}_{3,1,-1}[m_u^t, m_d^r] + \frac{1}{8} (g_1^6 + 2 g_1^4 g_2^2 - 3 g_1^2 g_2^4) \text{LF}_{3,1,-1}[\tilde{\mu}, m_1] - \\
& \frac{3}{16} (g_1^6 - 3 g_1^4 g_2^2) \text{LF}_{3,2,-2}[\tilde{\mu}, m_1] - \frac{3}{16} g_1^4 m_1^2 (g_1^2 - 3 g_2^2) \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] + \\
& \frac{1}{24} (-6 g_1^6 - 7 g_1^4 g_2^2 + 13 g_1^2 g_2^4) \text{LF}_{4,1,-2}[\tilde{\mu}, m_1] + \frac{1}{8} (g_1^6 - 3 g_1^4 g_2^2) \text{LF}_{4,2,-3}[\tilde{\mu}, m_1] + \\
& \frac{1}{8} g_1^4 m_1^2 (g_1^2 - 3 g_2^2) \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{24} (3 g_1^6 + 2 g_1^4 g_2^2 - 5 g_1^2 g_2^4) \text{LF}_{5,1,-3}[\tilde{\mu}, m_1] + \\
& \frac{1}{24} (9 g_1^4 g_2^2 + 2 g_1^2 g_2^4 - 11 g_2^6) \text{LF}_{3,1,-1}[\tilde{\mu}, m_2] - \frac{5}{48} (17 g_1^2 g_2^4 - 35 g_2^6) \text{LF}_{3,2,-2}[\tilde{\mu}, m_2] - \\
& \frac{3}{16} g_2^4 m_2^2 (g_1^2 - 3 g_2^2) \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] - \frac{3}{8} (2 g_1^4 g_2^2 + g_1^2 g_2^4 - 3 g_2^6) \text{LF}_{4,1,-2}[\tilde{\mu}, m_2] + \\
& \frac{1}{8} g_2^4 (7 g_1^2 - 17 g_2^2) \text{LF}_{4,2,-3}[\tilde{\mu}, m_2] + \frac{1}{8} (g_2^4 m_2^2 (g_1^2 - 3 g_2^2) - 4 g_2^6 \tilde{\mu}^2) \text{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \\
& \frac{1}{8} (3 g_1^4 g_2^2 + 2 g_1^2 g_2^4 - 5 g_2^6) \text{LF}_{5,1,-3}[\tilde{\mu}, m_2] + (-g_1^4 g_2^2 + g_1^2 g_2^4) \text{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] + \\
& \frac{3}{4} g_1^2 g_2^2 (m_1 m_2 (-g_1^2 + g_2^2) + 4 g_2^2 \tilde{\mu}^2) \text{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
& \frac{1}{2} g_1^2 g_2^2 (g_1^2 - g_2^2) \text{LF}_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] + \frac{1}{2} g_1^2 g_2^2 (m_1 m_2 (g_1^2 - g_2^2) - 4 g_2^2 \tilde{\mu}^2) \\
& \text{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^{PR}} \overline{y_d^{ST}} y_d^{PT} y_d^{SR} \text{LF}_{2,1,1,0}[m_d^r, m_d^t, m_q^S] + \\
& \frac{1}{2} g_1^2 \overline{y_u^{ST}} y_u^{PT} \overline{a_d^{PR}} a_d^{SR} \text{LF}_{2,1,1,0}[m_d^r, m_q^P, m_q^S] + \frac{1}{2} g_1^2 y_d^{SR} \overline{y_u^{ST}} (\overline{a_d^{PR}} a_u^{PT} - \tilde{\mu}^2 \overline{y_d^{PR}} y_u^{PT}) \\
& \text{LF}_{2,1,1,0}[m_d^r, m_q^P, m_u^t] - \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} \text{LF}_{2,1,1,0}[m_d^r, m_q^S, m_u^t] + \\
& 3 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{UV} y_u^{PV} y_u^{UT} \text{LF}_{1,1,1,0}[m_d^r, m_u^t, m_u^V] + \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^{PR}} \overline{y_d^{ST}} y_d^{PT} y_d^{SR} \\
& \text{LF}_{2,1,1,0}[m_d^t, m_d^r, m_q^S] + \frac{1}{2} g_1^2 \overline{y_d^{ST}} y_u^{SR} a_d^{PT} \overline{a_u^{PR}} \text{LF}_{2,1,1,0}[m_d^t, m_q^P, m_u^r] + \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_e^{PR}} \overline{y_e^{ST}} y_e^{PT} y_e^{SR} \text{LF}_{2,1,1,0}[m_e^r, m_e^t, m_l^S] + \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_e^{PR}} \overline{y_e^{ST}} y_e^{PT} y_e^{SR} \\
& \text{LF}_{2,1,1,0}[m_e^t, m_e^r, m_l^S] - \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_e^{PR}} \overline{y_e^{ST}} y_e^{PT} y_e^{SR} \text{LF}_{2,1,1,0}[m_l^S, m_e^r, m_e^t] + \\
& \frac{1}{4} \tilde{\mu}^2 \overline{y_e^{PR}} \overline{y_e^{ST}} y_e^{PT} y_e^{SR} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_l^S, m_e^r, m_e^t] + \\
& \frac{1}{2} \overline{y_u^{ST}} y_u^{PT} \overline{a_d^{PR}} a_d^{SR} (2 g_1^2 - 3 g_2^2) \text{LF}_{2,1,1,0}[m_q^P, m_d^r, m_q^S] + \\
& \frac{3}{4} \overline{y_u^{ST}} y_u^{PT} \overline{a_d^{PR}} a_d^{SR} (-g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^P, m_d^r, m_q^S] + \\
& \frac{1}{2} y_d^{SR} \overline{y_u^{ST}} (\overline{a_d^{PR}} a_u^{PT} (2 g_1^2 - 3 g_2^2) + g_1^2 \tilde{\mu}^2 \overline{y_d^{PR}} y_u^{PT}) \text{LF}_{2,1,1,0}[m_q^P, m_d^r, m_u^t] - \\
& \frac{3}{4} y_d^{SR} \overline{y_u^{ST}} (\overline{a_d^{PR}} a_u^{PT} (g_1^2 - g_2^2) + \tilde{\mu}^2 \overline{y_d^{PR}} y_u^{PT} (g_1^2 + g_2^2)) \text{LF}_{3,1,1,-1}[m_q^P, m_d^r, m_u^t] + \\
& \frac{1}{2} \overline{y_d^{ST}} y_u^{SR} a_d^{PT} \overline{a_u^{PR}} (2 g_1^2 - 3 g_2^2) \text{LF}_{2,1,1,0}[m_q^P, m_d^t, m_u^r] + \\
& \frac{3}{4} \overline{y_d^{ST}} y_u^{SR} a_d^{PT} \overline{a_u^{PR}} (-g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^P, m_d^t, m_u^r] - \\
& \frac{3}{4} \overline{y_u^{ST}} y_u^{SR} a_d^{PT} \overline{a_u^{PR}} (-g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^P, m_d^t, m_u^r] - \\
& \frac{3}{8} \overline{y_u^{ST}} y_u^{PT} \overline{a_u^{PR}} a_u^{SR} (-g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_q^P, m_u^r, m_q^S] - \\
& \frac{1}{4} \overline{y_u^{ST}} y_u^{SR} \overline{a_u^{PR}} a_u^{PT} (g_1^2 + 3 g_2^2) \text{LF}_{2,1,1,0}[m_q^P, m_u^r, m_u^t] - \\
& \frac{3}{4} y_d^{SR} \overline{y_u^{ST}} (\overline{a_d^{PR}} a_u^{PT} (g_1^2 - g_2^2) + \tilde{\mu}^2 \overline{y_d^{PR}} y_u^{PT} (g_1^2 + g_2^2)) \text{LF}_{2,2,1,-1}[m_q^P, m_u^t, m_d^r] - \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^{PR}} \overline{y_d^{ST}} y_d^{PT} y_d^{SR} \text{LF}_{2,1,1,0}[m_q^S, m_d^r, m_d^t] + \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{PR}} \overline{y_d^{ST}} y_d^{PT} y_d^{SR} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^S, m_d^r, m_d^t] + \\
& \frac{1}{2} \overline{y_u^{ST}} y_u^{PT} \overline{a_d^{PR}} a_d^{SR} (2 g_1^2 - 3 g_2^2) \text{LF}_{2,1,1,0}[m_q^S, m_d^r, m_q^P] + \\
& \frac{3}{4} \overline{y_u^{ST}} y_u^{PT} \overline{a_d^{PR}} a_d^{SR} (-g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^S, m_d^r, m_q^P] + \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} \text{LF}_{2,1,1,0}[m_q^S, m_d^r, m_u^t] - \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^S, m_d^r, m_u^t] + \\
& \frac{1}{4} \overline{a_u^{PR}} a_u^{SR} (g_1^2 - 3 g_2^2) (\overline{y_d^{ST}} y_d^{PT} + 2 \overline{y_u^{ST}} y_u^{PT}) \text{LF}_{2,1,1,0}[m_q^S, m_q^P, m_u^r] + \\
& \frac{3}{4} \overline{y_u^{ST}} y_u^{PT} \overline{a_u^{PR}} a_u^{SR} (-g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^S, m_q^P, m_u^r] + \\
& \frac{3}{8} \overline{y_u^{ST}} y_u^{PT} \overline{a_u^{PR}} a_u^{SR} (-g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_q^S, m_u^r, m_q^S] + \\
& \frac{1}{4} \overline{y_u^{ST}} y_u^{SR} \overline{a_u^{PR}} a_u^{PT} (g_1^2 + 3 g_2^2) \text{LF}_{2,1,1,0}[m_q^P, m_u^r, m_u^t] - \\
& \frac{3}{4} y_d^{SR} \overline{y_u^{ST}} (\overline{a_d^{PR}} a_u^{PT} (g_1^2 - g_2^2) + \tilde{\mu}^2 \overline{y_d^{PR}} y_u^{PT} (g_1^2 + g_2^2)) \text{LF}_{2,2,1,-1}[m_q^P, m_u^t, m_d^r] - \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^{PR}} \overline{y_d^{ST}} y_d^{PT} y_d^{SR} \text{LF}_{2,1,1,0}[m_q^S, m_d^r, m_d^t] + \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{PR}} \overline{y_d^{ST}} y_d^{PT} y_d^{SR} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^S, m_d^r, m_d^t] + \\
& \frac{1}{2} \overline{y_u^{ST}} y_u^{PT} \overline{a_d^{PR}} a_d^{SR} (2 g_1^2 - 3 g_2^2) \text{LF}_{2,1,1,0}[m_q^S, m_d^r, m_q^P] + \\
& \frac{3}{4} \overline{y_u^{ST}} y_u^{PT} \overline{a_d^{PR}} a_d^{SR} (-g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_q^S, m_d^r, m_q^P] + \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} \text{LF}_{2,1,1,0}[m_q^S, m_d^r, m_u^t] - \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_q^S, m_u^t, m_d^r] - \\
& \frac{1}{2} \overline{y_d^{ST}} y_u^{SR} a_d^{PT} \overline{a_u^{PR}} (2 g_1^2 + 3 g_2^2) \text{LF}_{2,1,1,0}[m_u^r, m_d^t, m_q^P] + \\
& \frac{3}{2} g_2^2 \overline{y_d^{ST}} y_u^{SR} a_d^{PT} \overline{a_u^{PR}} \text{LF}_{3,1,1,-1}[m_u^r, m_d^t, m_q^P] - \\
& \frac{1}{4} \overline{a_u^{PR}} a_u^{SR} (\overline{y_d^{ST}} y_d^{PT} (7 g_1^2 + 3 g_2^2) + \overline{y_u^{ST}} y_u^{PT} (-7 g_1^2 + 3 g_2^2)) \text{LF}_{2,1,1,0}[m_u^r, m_q^P, m_q^S] + \\
& \frac{3}{4} \overline{a_u^{PR}} a_u^{SR} (\overline{y_d^{ST}} y_d^{PT} (g_1^2 + g_2^2) + \overline{y_u^{ST}} y_u^{PT} (-g_1^2 + g_2^2)) \text{LF}_{3,1,1,-1}[m_u^r, m_q^P, m_q^S] + \\
& \frac{1}{4} \overline{y_u^{ST}} y_u^{SR} \overline{a_u^{PR}} a_u^{PT} (7 g_1^2 - 3 g_2^2) \text{LF}_{2,1,1,0}[m_u^r, m_q^P, m_u^t] + \\
& \frac{3}{4} \overline{y_u^{ST}} y_u^{SR} \overline{a_u^{PR}} a_u^{PT} (-g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_u^r, m_q^P, m_u^t] - \\
& \frac{1}{4} \tilde{\mu}^2 \overline{y_u^{PR}} \overline{y_u^{ST}} y_u^{PT} y_u^{SR} (g_1^2 + 3 g_2^2) \text{LF}_{2,1,1,0}[m_u^r, m_q^S, m_u^t] + \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_u^{PR}} \overline{y_u^{ST}} y_u^{PT} y_u^{SR} (-g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_u^r, m_q^S, m_u^t] + \\
& \frac{3}{4} \overline{y_u^{ST}} y_u^{SR} \overline{a_u^{PR}} a_u^{PT} (-g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_u^r, m_u^t, m_q^P] + \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_u^{PR}} \overline{y_u^{ST}} y_u^{PT} y_u^{SR} (-g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_u^r, m_u^t, m_q^S] - \\
& \frac{1}{2} y_d^{SR} \overline{y_u^{ST}} (2 g_1^2 + 3 g_2^2) (\overline{a_d^{PR}} a_u^{PT} - \tilde{\mu}^2 \overline{y_d^{PR}} y_u^{PT}) \text{LF}_{2,1,1,0}[m_u^t, m_d^r, m_q^P] + \\
& \frac{3}{2} g_2^2 y_d^{SR} \overline{y_u^{ST}} (\overline{a_d^{PR}} a_u^{PT} - \tilde{\mu}^2 \overline{y_d^{PR}} y_u^{PT}) \text{LF}_{3,1,1,-1}[m_u^t, m_d^r, m_q^P] + \\
& \frac{1}{2} \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} (2 g_1^2 + 3 g_2^2) \text{LF}_{2,1,1,0}[m_u^t, m_d^r, m_q^S] - \\
& \frac{3}{2} g_2^2 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} \text{LF}_{3,1,1,-1}[m_u^t, m_d^r, m_q^S] + \\
& \frac{1}{4} \overline{y_u^{ST}} y_u^{SR} \overline{a_u^{PR}} a_u^{PT} (7 g_1^2 - 3 g_2^2) \text{LF}_{2,1,1,0}[m_u^t, m_q^P, m_u^r] + \\
& \frac{3}{4} \overline{y_u^{ST}} y_u^{SR} \overline{a_u^{PR}} a_u^{PT} (-g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_u^t, m_q^P, m_u^r] - \\
& \frac{1}{4} \tilde{\mu}^2 \overline{y_u^{PR}} \overline{y_u^{ST}} y_u^{PT} y_u^{SR} (g_1^2 + 3 g_2^2) \text{LF}_{2,1,1,0}[m_u^t, m_q^S, m_u^r] + \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_u^{PR}} \overline{y_u^{ST}} y_u^{PT} y_u^{SR} (-g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_u^t, m_q^S, m_u^r] + \\
& \frac{3}{8} g_1^2 g_2^2 (g_1^2 - g_2^2) \text{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_2] + \frac{1}{8} (-7 g_1^4 g_2^2 + g_1^2 g_2^4) \text{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_2] + \\
& \frac{3}{8} g_1^2 g_2^2 (-m_1 m_2 (g_1^2 + g_2^2) + 8 g_2^2 \tilde{\mu}^2) \text{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] + \frac{1}{4} g_1^4 g_2^2 \text{LF}_{3,2,1,-3}[\tilde{\mu}, m_1, m_2] + \\
& \frac{1}{4} g_1^4 g_2^2 (m_1^2 + 2 m_1 m_2 - 8 \tilde{\mu}^2) \text{LF}_{3,2,1,-2}[\tilde{\mu}, m_1, m_2] - 2 g_1^4 g_2^2 m_1^2 \tilde{\mu}^2 \text{LF}_{3,2,1,-1}[\tilde{\mu}, m_1, m_2] + \\
& \frac{1}{2} g_1^4 g_2^2 \text{LF}_{4,1,1,-3}[\tilde{\mu}, m_1, m_2] + \frac{1}{4} g_1^2 g_2^2 (m_1 m_2 (g_1^2 + g_2^2) - 10 g_2^2 \tilde{\mu}^2) \text{LF}_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] + \\
& \frac{1}{4} g_1^2 g_2^2 (2 g_1^2 - g_2^2) \text{LF}_{3,2,1,-3}[\tilde{\mu}, m_2, m_1] + \\
& \frac{1}{4} g_1^2 g_2^2 (2 m_1 m_2 g_1^2 + g_2^2 (m_2^2 - 24 \tilde{\mu}^2)) \text{LF}_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] + \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^{ST}} y_d^{SR} (\overline{a_d^{PR}} a_d^{PT} - \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PT}) \text{LF}_{2,1,1,0}[m_d^r, m_d^t, m_q^P, m_q^S] + \\
& 3 \tilde{\mu}^2 \overline{y_d^{PR}} \overline{y_d^{ST}} y_d^{PT} y_d^{UR} \overline{y_u^{UV}} y_u^{SV} \text{LF}_{1,1,1,0}[m_d^r, m_d^t, m_q^P, m_u^V] + \\
& 3 \overline{y_u^{ST}} \overline{y_u^{UV}} y_u^{PV} y_u^{UT} \overline{a_d^{PR}} a_d^{SR} \text{LF}_{1,1,1,0}[m_d^r, m_q^P, m_q^S, m_q^U] + \\
& \frac{1}{2} g_1^2 (\tilde{\mu}^4 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} + \overline{a_d^{PR}} a_u^{PT} (a_d^{SR} \overline{a_u^{ST}} - \tilde{\mu}^2 y_d^{SR} \overline{y_u^{ST}})) \\
& \text{LF}_{2,1,1,0}[m_d^r, m_q^P, m_q^S, m_u^t] + 3 y_d^{UR} \overline{y_u^{ST}} \overline{y_u^{UV}} y_u^{PT} \overline{a_d^{PR}} a_u^{SV} \text{LF}_{1,1,1,0}[m_d^r, m_q^P, m_q^S, m_u^V] + \\
& 3 \overline{y_u^{ST}} \overline{y_u^{UV}} (y_d^{UR} y_u^{SV} \overline{a_d^{PR}} a_u^{PT} - \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{SR} y_u^{PV} y_u^{UT}) \text{LF}_{1,1,1,0}[m_d^r, m_q^P, m_u^t, m_u^V] - \\
& 3 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} \overline{y_u^{UV}} y_u^{PV} y_u^{UT} \text{LF}_{1,1,1,0}[m_d^r, m_q^S, m_u^t, m_u^V] + \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^{ST}} y_d^{SR} \overline{a_d^{PR}} a_d^{PT} \text{LF}_{2,1,1,0}[m_d^t, m_q^P, m_q^S] - \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^{ST}} y_u^{SR} a_d^{PT} \overline{a_u^{PR}} \text{LF}_{2,1,1,0}[m_d^t, m_q^P, m_q^S, m_u^r] + \\
& 3 \overline{y_d^{ST}} \overline{y_u^{UV}} y_u^{PV} y_u^{SR} a_d^{UT} \overline{a_u^{PR}} \text{LF}_{1,1,1,0}[m_d^t, m_q^P, m_q^U, m_u^r] + \\
& 3 \overline{y_d^{ST}} \overline{y_u^{UV}} \overline{a_u^{PR}} (y_d^{UT} y_u^{SR} a_u^{PV} + y_u^{SV} y_u^{UR} a_d^{PT}) \text{LF}_{1,1,1,0}[m_d^t, m_q^P, m_u^r, m_u^V] + \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_e^{ST}} y_e^{SR} (\overline{a_e^{PR}} a_e^{PT} - \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PT}) \text{LF}_{2,1,1,0}[m_e^r, m_e^t, m_l^P, m_l^S] + \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_e^{ST}} y_e^{SR} \overline{a_e^{PR}} a_e^{PT} \text{LF}_{2,1,1,0}[m_e^t, m_e^r, m_l^P, m_l^S] + \\
& \frac{1}{2} \tilde{\mu}^2 \overline{y_e^{ST}} y_e^{SR} (-g_2^2 \overline{a_e^{PR}} a_e^{PT} + g_1^2 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PT}) \text{LF}_{2,1,1,0}[m_l^P, m_e^r, m_e^t, m_l^S] - \\
& \frac{1}{4} \tilde{\mu}^2 \overline{y_e^{ST}} y_e^{SR} (g_1^2 - g_2^2) (\overline{a_e^{PR}} a_e^{PT} + \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PT}) \text{LF}_{3,1,1,1,-1}[m_l^P, m_e^r, m_e^t, m_l^S] - \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_e^{ST}} y_e^{SR} \overline{a_e^{PR}} a_e^{PT} \text{LF}_{2,1,1,0}[m_l^S, m_e^r, m_e^t, m_l^P] + \\
& \frac{1}{4} \tilde{\mu}^2 \overline{y_e^{ST}} y_e^{SR} \overline{a_e^{PR}} a_e^{PT} (g_1^2 + g_2^2) \text{LF}_{3,1,1,1,-1}[m_l^S, m_e^r, m_e^t, m_l^P] + \\
& \frac{1}{2} \tilde{\mu}^2 \$$